Chapter 5: Parametric regression models — Exercises

STA310 Survival Analysis (Master's in Applied Biostatistics)

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Chapter 5 introduces models that assume a specific probability distribution for the survival times. These exercises cover the common distributions (exponential, Weibull, log-logistic, lognormal, Gompertz), how to assess their suitability, the Weibull proportional hazards model, the accelerated failure time model, and the proportional odds model.

1 Distributions for survival times (Book §5.1)

1.1 The Weibull distribution

Type: Concept & notation

The Weibull hazard function is $h(t) = \lambda\gamma t^{\gamma-1}$ for $\lambda, \gamma > 0$.

(a) Which parameter is the *shape* parameter and which is the *scale* parameter, and how does each affect the hazard function?

(b) What special case is obtained when $\gamma = 1$?

(c) State, in words, the shape of the hazard function for $\gamma < 1$ and for $\gamma > 1$. What limitation of the Weibull hazard does this reveal?

1.2 Hazard shapes of the common distributions

Type: Method selection

Match each situation to the most appropriate distribution, and justify briefly: exponential, Weibull, log-logistic, lognormal, Gompertz.

(a) The risk of the event is believed to be roughly constant over the follow-up period.
(b) The risk is expected to rise steadily over time (e.g. mortality increasing with age).
(c) The risk is expected to rise shortly after an intervention and then fall (e.g. death following a heart transplant, as the patient recovers).

2 Choosing and fitting a parametric model (Book §5.2–5.4)

2.1 The log-cumulative hazard plot

Type: Interpret figure

Builds on: Chapter 2

For a Weibull distribution, $\log\{-\log S(t)\} = \log\lambda + \gamma \log t$.

(a) How is the plot constructed, and what does a straight line indicate?
(b) What do the slope and intercept of the line estimate?
(c) You obtain a straight line with slope close to 1. What does this suggest, and what model would you then consider?

2.2 Fitting the exponential distribution

Type: Concept & notation

For a single sample from an exponential distribution with r events and total follow-up time $\sum_i t_i$, the maximum likelihood estimate is $\hat{\lambda} = r / \sum_i t_i$.

In a study of 18 women using an intrauterine device, 9 discontinued use and the total observed time was 1046 days.

(a) Compute $\hat{\lambda}$, and state the estimated hazard and survivor functions.
(b) Compute the estimated median discontinuation time, and the estimated 90th percentile.
(c) The standard error of $\hat{\lambda}$ is $\hat{\lambda}/\sqrt{r}$. Explain in words why the precision of $\hat{\lambda}$ depends on the number of events, not the number of individuals.

2.3 Choosing between parametric and Cox models

Type: Model comparison

(a) State one advantage and one disadvantage of a Weibull proportional hazards model relative to the Cox model.
(b) When a Weibull model is a reasonable description of the data, what happens to the standard errors of the estimated coefficients compared with the Cox model?

(c) In practice, how might you decide between the two when there is little to choose in goodness of fit?

3 The Weibull proportional hazards model (Book §5.5–5.9)

3.1 The Weibull proportional hazards model

Type: Concept & notation

Builds on: Chapter 3

The Weibull proportional hazards model is $h_i(t) = \exp(\beta_1 x_{1i} + \cdots + \beta_p x_{pi})\, \lambda\gamma t^{\gamma-1}$.

(a) What is the baseline hazard function, and what distribution do the survival times follow?
(b) How do the explanatory variables affect the distribution of survival times — through the scale or the shape parameter?
(c) How are the β-coefficients interpreted, and what is the "proportional hazards property" of the Weibull distribution?

3.2 Interpreting Weibull output

Type: Interpret R output

A Weibull model is fitted (in its log-linear form, $\log T_i = \mu + \alpha x_i + \sigma\epsilon_i$) to the breast cancer data, where $x = 1$ for positively stained and 0 for negatively stained tumours. The output gives $\hat{\alpha} = -0.997$ and $\hat{\sigma} = 1.067$ (reported as `Scale`).

(a) What is the estimated Weibull shape parameter γ?
(b) What does $\exp(\hat{\alpha}) = \exp(-0.997) \approx 0.37$ mean, in words?
(c) Using $\beta = -\alpha/\sigma$, compute the estimated log hazard ratio and hazard ratio for positive versus negative staining.

4 The accelerated failure time model (Book §5.10–5.13)

4.1 The time ratio and the accelerated failure time model

Type: Concept & notation

For two groups the accelerated failure time (AFT) model is $S_N(t) = S_S(t/\phi)$.

(a) What is the parameter ϕ called, and what does $\phi < 1$ mean in terms of the time to the event?
(b) Relate ϕ to the median survival times in the two groups.
(c) If $\phi = e^{\alpha}$ and $\hat{\alpha} = -0.4$, interpret the effect of the new treatment on survival time.

4.2 Accelerated failure time versus proportional hazards

Type: Model comparison

(a) In words, how do the AFT and proportional hazards models differ in the way covariates act?
(b) Which single distribution possesses *both* the proportional hazards and the accelerated failure time properties, and what does this imply?
(c) In what situations might an AFT model be preferable to a proportional hazards model?

5 Other parametric models (Book §5.15–5.16)

5.1 The proportional odds model

Type: Concept & notation

In the proportional odds model, $\frac{S_i(t)}{1-S_i(t)} = e^{\eta_i}\frac{S_0(t)}{1-S_0(t)}$.

(a) What quantity do the explanatory variables act on, and how are the coefficients interpreted?
(b) What important property does the hazard ratio $h_i(t)/h_0(t)$ have in this model, and in what kind of applications is this useful?
(c) Which distribution has the proportional odds property (and also the AFT property)?

6 Putting it together

6.1 A misleading interpretation

Type: What's wrong here?

A researcher fits a Weibull accelerated failure time model and obtains a time ratio of 0.5 for the new treatment. The researcher concludes: "The new treatment halves the risk of death."

Explain what is wrong with this conclusion.

7 Mini-lab: fitting parametric models in R

7.1 Fitting a Weibull model with `survreg`

Type: Mini-lab

Run the code below. It simulates 300 patients in two treatment groups with Weibull survival times; the treated group has its survival times multiplied by 0.5 (an accelerated failure time effect). A Weibull model is fitted with `survreg`.

```
set.seed(911)
n <- 300
treat <- rep(c(0, 1), each = n/2)
shape <- 0.8
scale0 <- 20
t0 <- rweibull(n/2, shape = shape, scale = scale0)
t1 <- 0.5 * rweibull(n/2, shape = shape, scale = scale0)  # time ratio 0.5
t <- c(t0, t1)
cens <- runif(n, 0, 80)
obs <- pmin(t, cens)
status <- as.numeric(t <= cens)
d <- data.frame(time = obs, status = status, treat = treat)

library(survival)
```

```r
fit <- survreg(Surv(time, status) ~ treat, data = d, dist = "weibull")
summary(fit)
```

```
Call:
survreg(formula = Surv(time, status) ~ treat, data = d, dist = "weibull")
              Value Std. Error     z       p
(Intercept)  3.0312     0.1201 25.23 < 2e-16
treat       -0.6685     0.1603 -4.17 3.1e-05
Log(scale)   0.1933     0.0528  3.66 0.00026

Scale= 1.21

Weibull distribution
Loglik(model)= -857.1   Loglik(intercept only)= -865.7
    Chisq= 17.27 on 1 degrees of freedom, p= 3.2e-05
Number of Newton-Raphson Iterations: 5
n= 300
```

```r
exp(coef(fit))
```

```
(Intercept)       treat
 20.7228810   0.5124987
```

(a) What is the estimated coefficient of `treat`, and what does `exp(coef(fit))` give for `treat`? Interpret this quantity.

(b) The output reports `Scale`. What is the estimated Weibull shape parameter, and what does it say about the hazard over time?

(c) The coefficient of `treat` is on the log-time scale. What is the corresponding hazard ratio under the Weibull proportional hazards model? Is the treatment beneficial?